

Notes: Li, Liu, & Wei (MS, 2026)

Credit Rating Purchases and S&P 500 Index Membership Decisions

Contents

1	Big picture	3
2	Institutional setting and economic motivation	4
2.1	Why S&P 500 additions are valuable	4
2.2	Why the S&P 500 is not a purely mechanical top-500 list	4
2.3	Why credit ratings create a governance issue	4
3	Data and measurement	5
3.1	Fund-market data and benchmark growth	5
3.2	Index membership and addition data	5
3.3	Coding S&P's published selection methodology	5
3.4	Credit-rating purchase data	6
3.5	Main variables	6
4	Models and empirical specifications	7
4.1	Rule-prediction exercise	7
4.2	Published-rule prediction	7
4.3	Baseline addition-probability regression	7
4.4	Decomposing purchase types	7
4.5	Agency-specific comparison	7
4.6	Size-rank heterogeneity	8
4.7	Random forest horse race	8
4.8	Vacancy test: do firms buy S&P ratings when openings appear?	8
4.9	Payoff test: do firms buy more when the inclusion prize is higher?	8
4.10	2002 foreign-firm rule-change event study	9
4.11	Discretionary entrant performance	9
5	Identification and empirical design	10
5.1	Main identifying comparison	10
5.2	Why Moody's is not just another control	10
5.3	Why vacancy events improve credibility	10
5.4	Why the 2002 rule change is especially informative	10
5.5	Important alternative explanations	11
5.6	Summary: identification and empirical design	11
6	Tables and figures	12
6.1	Table 1 + Figure 1: The benchmark ecosystem becomes large	12
6.2	Table 2 + Table 3: How much discretion is there?	12
6.3	Table 4 + Figure 2: Rating purchases and addition probability	13
6.4	Table 5 + Table 6: Do firms strategically buy more S&P ratings?	13
6.5	Table 7: 2002 foreign-firm rule change	15
6.6	Figure 3 + Table 8: Are discretionary entrants better firms?	16

7	What the Internet Appendix adds	17
8	Interpretation	18
8.1	What is the strongest evidence?	18
8.2	What is the most important caveat?	18
8.3	Relation to index-addition literature	18
8.4	Relation to credit-rating literature	19
8.5	Relation to conflict-of-interest literature in financial institutions	19
8.6	Relation to benchmark governance	20
8.7	Relation to post-addition performance literature	20
8.8	Overall contribution relative to related literature	20
9	Short summary	21
10	Conclusion	22
10.1	Core conclusion	22
10.2	Economic intuition	22
10.3	Incremental contribution	22
10.4	Takeaway	22

1 Big picture

- **Question.** Is S&P 500 membership determined by objective, published index methodology, or does the index provider use discretion in a way that is correlated with firms' purchases of S&P credit ratings?
- **Why the question matters.** The S&P 500 is a benchmark used by passive index funds, active benchmarked funds, derivatives markets, asset-pricing research, the VIX, and macro-finance applications.
- **Economic value of membership.** A firm added to the S&P 500 can benefit from index-fund demand, increased visibility, prestige, and potentially a lower cost of capital. This creates a prize that firms may have incentives to pursue.
- **Core conflict-of-interest concern.** S&P Global both operates the S&P 500 index and sells credit ratings. Candidate firms may believe that buying S&P ratings improves their chance of index inclusion, even though rating purchases are not listed as an index-selection criterion.
- **First empirical fact: discretion.** The authors code S&P's published methodology and show that the rules explain much of membership status but only a limited part of addition decisions. Roughly one in five additions violates at least one coded criterion, and many eligible firms are not selected.
- **Second empirical fact: S&P-specific rating purchases.** Recent purchases of S&P ratings predict index addition, while recent Moody's rating purchases do not show the same pattern.
- **Magnitude.** The baseline OLS coefficient of 0.214 means that a recent S&P rating purchase raises the probability of addition by about 0.214 percentage points. The unconditional quarterly probability of addition is only about 0.155 percentage points.
- **Why this is economically large.** Because addition events are rare, a coefficient of roughly 0.2 percentage points is large relative to the base rate. It is not just statistically significant; it is economically meaningful.
- **Nonparametric horse race.** A random forest ranks recent S&P rating purchases above many variables explicitly listed in S&P's public methodology.
- **Strategic-purchase evidence.** Firms buy more S&P ratings when mergers between existing S&P 500 firms create vacancies in the index. The effect is concentrated among firms with plausible size ranks for addition.
- **Payoff evidence.** Rating purchases rise more when the expected stock-price reward from index inclusion is higher, proxied by high recent addition announcement returns.
- **Rule-change evidence.** In 2002, S&P changed the rule so that foreign firms were no longer eligible for S&P 500 membership. After the rule change, foreign firms reduce S&P rating purchases relative to U.S. firms, but not Moody's purchases.
- **Performance evidence.** If discretion were benign and based on superior information about firm quality, discretionary entrants might outperform. Instead, discretionary entrants perform worse after inclusion than rules-based entrants and matched discretionary outs.
- **Main interpretation.** The paper does not prove a direct quid pro quo. It documents several mutually reinforcing patterns that are difficult to reconcile with a purely transparent, rule-based index construction process.
- **Broad takeaway.** Index governance matters because benchmark construction allocates capital. When the same organization controls a benchmark and sells services to candidate firms, even perceived conflicts can have market-wide consequences.

2 Institutional setting and economic motivation

2.1 Why S&P 500 additions are valuable

- Passive funds mechanically buy index constituents; benchmarked active funds may also increase holdings to reduce tracking error.
- Index membership can create immediate demand pressure around announcement and effective dates.
- Inclusion can make a firm more visible to analysts, investors, and executives who use the index as a market proxy.
- Membership may affect firm-level outcomes because stock-demand effects can lower the cost of capital.
- Because entry is valuable and rare, firms near the index boundary have incentives to maximize their chances of being chosen.

Interpretation: The paper treats index membership as a valuable allocation decision. If index entry were not valuable, there would be little reason for firms to strategically purchase services from the index provider.

2.2 Why the S&P 500 is not a purely mechanical top-500 list

- The S&P 500 is commonly described as a large-cap U.S. stock index, but it is not simply the 500 largest stocks.
- S&P uses a committee process and a methodology that includes size, liquidity, profitability, public float, domicile, listing, and sector representation.
- Some criteria are precise, such as minimum liquidity or earnings conditions; other criteria are more qualitative, such as sector representation.
- The committee can select among many eligible firms and can sometimes waive criteria.
- Therefore, a firm can be eligible but not chosen, and a firm can be chosen even though one or more coded criteria are not satisfied.

Interpretation: Discretion is not automatically improper. The paper's concern is that discretion creates room for conflicts of interest when one business unit sells services to firms seeking favorable treatment from another business unit.

2.3 Why credit ratings create a governance issue

- S&P is one of the dominant credit rating agencies and uses an issuer-pay business model.
- Firms purchase ratings for issuer-level creditworthiness and for specific debt issues.
- New rating purchases are especially relevant because initiation fees are generally larger than renewal fees.
- The exact fee amounts are confidential and negotiated, so the paper cannot observe payments directly.
- The authors therefore use new rating purchases as an observable proxy for fee-generating interactions between firms and S&P.
- Moody's serves as the key comparison because it is a close substitute in the ratings market but does not control S&P 500 membership.

Interpretation: The empirical design is strongest when it shows that S&P purchases behave differently from Moody's purchases. That difference is what links the rating-purchase pattern to index membership rather than to ordinary bond-market activity.

3 Data and measurement

3.1 Fund-market data and benchmark growth

- The paper uses Morningstar data to document the rise of index funds and benchmarked funds.
- The fund-market evidence covers all U.S. equity funds, funds passively tracking the S&P 500, other funds benchmarked to the S&P 500, and analogous Russell 1000 funds.
- By 2019, passive S&P 500 index funds have about \$1.85 trillion in AUM.
- Other funds benchmarked to the S&P 500 add another roughly \$1.83 trillion.
- The S&P 500-related fund ecosystem reaches trillions of dollars and a large share of U.S. GDP.

Interpretation: This section motivates the entire paper. The larger the pool of assets linked to the benchmark, the more valuable the right to enter the benchmark becomes.

3.2 Index membership and addition data

- The index-addition sample covers 1980–2018.¹
- The authors identify S&P 500 additions using Sibilis Research and validate constituent information using CRSP.
- They exclude secondary share-class additions and spin-offs of existing member firms because these are not ordinary discretionary addition decisions.
- The final addition sample contains 773 additions.
- The main firm-quarter sample used for addition-probability regressions contains 479,101 observations.
- The dependent variable is scaled as $SP_Add_{i,t} = 100$ when firm i is added in quarter t and zero otherwise.

Interpretation: Scaling the dependent variable to 100 makes linear-probability coefficients easy to read as percentage-point changes.

3.3 Coding S&P’s published selection methodology

The authors translate S&P’s published methodology into observable variables.² The main criteria include:

- **Domicile and exchange listing:** the firm must satisfy U.S. domicile and major-exchange listing requirements.
- **Market capitalization:** the firm must exceed the relevant S&P 500 market-cap threshold.
- **Liquidity:** turnover must be at least one, and monthly trading volume must be at least 250,000 shares.
- **Financial viability:** recent quarterly earnings and trailing four-quarter earnings must be positive.
- **Public float:** the investable weight factor must exceed the required threshold.
- **Sector representation:** the committee considers how candidate firms affect the index’s sector balance.

¹Index ecosystem: Appendix Table IA1 names the leading S&P 500 funds and shows that by 2019 the top vehicles include Vanguard 500 Index Admiral, SPDR S&P 500 ETF Trust, Fidelity 500 Index, and iShares Core S&P 500 ETF. This makes the economic stakes concrete: index membership affects a multi-trillion-dollar ecosystem of tracking and benchmarking capital.

²Rule coding and historical thresholds: The paper codes S&P’s published methodology into observable criteria: exchange/common stock eligibility, domicile, market capitalization, turnover, monthly trading volume, financial viability, public float/IWF, sector representation, and post-2017 exceptions for S&P 400/600 firms. Appendix Table IA2 lists historical market-cap thresholds, including \$5.0bn in 2007, \$3.0bn in late 2008, \$4.0bn in 2011, \$6.1bn in 2017, and \$8.2bn in 2019.

- **Explicit exceptions:** when the methodology states that rules are waived for certain situations, the authors code those exceptions as part of the rule system rather than as discretion.

Interpretation: The paper’s measure of discretion is conservative in spirit: if S&P publicly explains an exception, the authors try to treat that exception as rules-based.

3.4 Credit-rating purchase data

- S&P ratings are obtained from Capital IQ/S&P Credit Rating data through WRDS.³
- Moody’s ratings are obtained from Moody’s’s Rating Delivery Services historical database, supplemented in limited cases by Mergent FISD.
- The paper observes issuer-level ratings and issue-level ratings.
- Unsolicited ratings are excluded because the mechanism concerns purchases by firms.
- The main variables focus on new ratings rather than annual renewals.
- The main purchase window is quarter t through $t - 4$, a five-quarter recent-purchase window.

Interpretation: The five-quarter window is designed to capture recent interactions with the rating agency, not the permanent fact that a firm has ever been rated.

3.5 Main variables

- $SP_Add_{i,t}$: equals 100 if firm i is added to the S&P 500 in quarter t , zero otherwise.
- $Purchase_SP_{i,t-4,t}$: equals one if firm i purchased a new S&P rating in quarter t or any of the previous four quarters.
- $Purchase_Moody_{i,t-4,t}$: equals one for analogous Moody’s purchases.
- $Purchase_Any_{i,t-4,t}$: equals one if the firm purchased from either S&P or Moody’s.
- $Purchase_Both$: equals one if the firm purchased new ratings from both agencies.
- $Purchase_SPOnly$ and $Purchase_MoodyOnly$: distinguish agency-specific purchases.
- $Bond_{i,t}$: controls for bond issuance because bond issuance naturally creates demand for ratings.
- $SPmerger_t$: marks quarters in which mergers between current S&P 500 firms create vacancies.
- $HighCAR_t$: identifies periods when the recent stock-price payoff from index addition is high.

Interpretation: The core empirical logic is to isolate the S&P-specific component of rating purchases from generic rating demand.

³Credit-rating purchases: The key purchase variables are new ratings from S&P and Moody’s during the current quarter and previous four quarters. The appendix adds robustness using FISD ratings and alternative purchase-timing specifications.

4 Models and empirical specifications

4.1 Rule-prediction exercise

The first step asks how much of index membership and index addition can be explained by public methodology variables:

$$IndexStatus_{i,t} = f(Rules_{i,t}), \quad SP_Add_{i,t} = f(Rules_{i,t}).$$

Here $Rules_{i,t}$ includes size, liquidity, profitability, float, listing, domicile, and sector-representation variables.

Interpretation: If the index were close to a mechanical rule-based process, the same observable rules should predict additions well. The paper finds that they explain membership better than additions, which implies a meaningful role for committee choice at the addition stage.

4.2 Published-rule prediction

$$Y_{it} = \alpha_t + \beta' Rules_{it} + u_{it},$$

where Y_{it} is either S&P 500 membership or addition status. If the index is mechanically rule-based, the public rules should explain additions almost as well as membership.

4.3 Baseline addition-probability regression

The central specification is:

$$SP_Add_{i,t} = \beta_1 Purchase_SP_{i,t-4,t} + \beta_2 Purchase_Any_{i,t-4,t} + \gamma' C_{i,t-1} + F_i + X_t + \varepsilon_{i,t}. \quad (1)$$

- $C_{i,t-1}$ contains all published-methodology controls and additional firm controls.
- F_i are firm fixed effects, absorbing persistent firm differences.
- X_t are quarter fixed effects, absorbing aggregate rating-market and index-market conditions.
- $Purchase_Any$ controls for general rating activity; $Purchase_SP$ captures the incremental effect of buying from S&P rather than not.

Interpretation: The coefficient β_1 is the main object. A positive value says that recent S&P rating purchases matter even after controlling for the fact that the firm recently bought some rating at all.

4.4 Decomposing purchase types

$$SP_Add_{it} = \beta_B Purchase_Both + \beta_S Purchase_SPOnly + \beta_M Purchase_MoodyOnly + Controls + FE + \varepsilon_{it}.$$

This specification asks whether the effect is specifically tied to S&P, not just rating demand.

4.5 Agency-specific comparison

A second version estimates:

$$SP_Add_{i,t} = \beta_{SP} Purchase_SP_{i,t-4,t} + \beta_M Purchase_Moody_{i,t-4,t} + \gamma' C_{i,t-1} + F_i + X_t + \varepsilon_{i,t}. \quad (2)$$

The paper also separates purchases into:

$$Purchase_Both, \quad Purchase_SPOnly, \quad Purchase_MoodyOnly.$$

Interpretation: The cleanest pattern for the conflict-of-interest interpretation is $\beta_{SP} > 0$ and $\beta_M \approx 0$, especially when S&P-only purchases are positive and Moody's-only purchases are not.

4.6 Size-rank heterogeneity

The paper estimates interactions between *Purchase_SP* and size-rank groups:

$$Purchase_SP_{i,t-4,t} \times SizeRankGroup_{i,t}.$$

The size groups include firms ranked 1–300, 301–500, 501–700, and 701–1,000 by market capitalization.

Interpretation: A real index-membership motive should be stronger for firms near the plausible addition region. Very small firms are less likely to be considered for the S&P 500, so S&P purchases should matter less for them.

4.7 Random forest horse race

The paper uses a random forest to allow nonlinearities and interactions among candidate variables:

$$SP_Add_{i,t} = RF(X_{i,t}),$$

where $X_{i,t}$ includes public methodology variables and rating-purchase variables.

Interpretation: The random forest is not the main causal design. Its value is descriptive: it asks whether *Purchase_SP* is empirically important relative to official criteria in predicting additions.

4.8 Vacancy test: do firms buy S&P ratings when openings appear?

Mergers between current S&P 500 constituents create vacancies. The paper tests whether firms react by buying ratings:

$$Purchase_{i,t}^a = \beta_1 SPmerger_t + \sum_g \beta_g (SPmerger_t \times SizeGroup_{g,i,t}) + \theta Bond_{i,t} + \gamma' C_{i,t-1} + F_{i,a} + X_{t,a} + u_{i,a,t}, \quad (3)$$

where a indexes the rating agency in a stacked S&P/Moody's framework.

Interpretation: This reverses the direction of the evidence. The baseline regression says S&P purchases predict additions; the vacancy test asks whether possible additions trigger S&P purchases.

4.9 Payoff test: do firms buy more when the inclusion prize is higher?

The key interaction is:

$$SPmerger_t \times SizeGroup_{i,t} \times HighCAR_t. \quad (4)$$

- $HighCAR_t$ equals one when recent index-addition CARs are high.
- The logic is that a larger expected stock-price reward increases the payoff to competing for index entry.
- The comparison is again S&P versus Moody's purchases.

Interpretation: Strategic behavior should be stronger when the prize is larger. This is why the payoff interaction is an important mechanism test.

4.10 2002 foreign-firm rule-change event study

The rule-change specification is:

$$Purchase_{i,t}^a = \beta(Post_t \times Foreign_i) + \theta Bond_{i,t} + \gamma' Z_{i,t} + \mu_{i,a} + \lambda_{t,a} + e_{i,a,t}. \quad (5)$$

- The sample is centered on 2001Q3–2003Q3.
- $Post_t$ marks 2002Q3 and later.
- $Foreign_i$ identifies non-U.S. firms listed on major U.S. exchanges.
- The stacked agency framework allows the paper to compare S&P and Moody's directly.

Interpretation: Foreign firms lose the S&P 500 membership motive after the rule change. A decline in S&P purchases but not Moody's purchases is a strong signal that the rating-purchase motive is tied to the index.

4.11 Discretionary entrant performance

The paper compares three groups:

- **Rules-based entrants:** entrants whose selection is consistent with the published methodology.
- **Discretionary entrants:** entrants chosen despite weaker fundamentals or rule violations.
- **Discretionary outs:** eligible nonmembers that appear stronger than discretionary entrants but are not chosen.

Returns are measured around the announcement date and the effective date:

$$CAR_{g,\tau} = \sum_{s=0}^{\tau} r_{g,s},$$

with market-adjusted and DGTW-adjusted variants.

Interpretation: If discretion reflects superior hidden information, discretionary entrants should perform well after inclusion. Their weaker post-entry returns make that benign story less compelling.

5 Identification and empirical design

5.1 Main identifying comparison

- The main regression compares firms that recently bought S&P ratings with firms that did not, conditional on eligibility controls, firm fixed effects, and quarter fixed effects.
- The coefficient is not interpreted as a pure randomized causal effect.
- The paper instead builds credibility by showing consistent patterns across several designs.
- The most important comparison is S&P versus Moody's because both agencies sell ratings, but only S&P controls S&P 500 membership.

Interpretation: The identifying assumption is not that firms buy ratings randomly. The assumption is that after controls and Moody's comparisons, the S&P-specific residual pattern is informative about index-related incentives.

5.2 Why Moody's is not just another control

- Moody's captures generic reasons to buy ratings: debt issuance, expansion, visibility, financing needs, and credit-market access.
- Moody's does not control S&P 500 membership, so it should not be directly tied to index committee decisions.
- If a firm buys ratings because it is about to issue a bond, both S&P and Moody's variables should move together.
- If the S&P coefficient is stronger, then agency-specific incentives are likely involved.

Interpretation: The Moody's benchmark is the central placebo-like comparison throughout the paper.

5.3 Why vacancy events improve credibility

- A merger between two existing S&P 500 constituents creates a mechanical need for at least one replacement.
- Nonmember firms do not choose when existing members merge.
- If potential entrants respond to these vacancies by buying S&P ratings, the behavior suggests forward-looking strategic incentives.
- The size-rank heterogeneity strengthens this logic because plausible candidates respond more than implausible candidates.

Interpretation: Vacancy events are not used to prove that S&P rewards purchases. They show that firms act as if purchases may help when an opportunity appears.

5.4 Why the 2002 rule change is especially informative

- The 2002 rule change removed foreign firms' eligibility for S&P 500 membership.
- It changed the index-related benefit of S&P rating purchases for foreign firms, while leaving many ordinary credit-market reasons to buy ratings intact.
- A decline in S&P purchases by foreign firms is therefore difficult to explain using generic debt-financing demand.
- The absence of a similar Moody's decline strengthens the interpretation.

Interpretation: This is the closest evidence to a quasi-experimental shock to the index-membership motive.

5.5 Important alternative explanations

- **Benign private information.** S&P may learn information during the rating process and use it to pick better firms for the index.
- **Firm visibility.** Firms buying S&P ratings may simply become more visible or investable.
- **Debt issuance.** Rating purchases may proxy for bond issuance and financial maturity.
- **Methodology coding error.** The authors may not perfectly implement S&P’s methodology, especially when criteria are qualitative.
- **Unobserved firm events.** Some corporate actions may simultaneously lead firms to buy ratings and make them attractive index candidates.

Interpretation: The paper’s design does not eliminate every alternative explanation. It narrows them by using Moody’s comparisons, vacancy timing, payoff interactions, foreign-firm rule changes, and post-inclusion performance.

5.6 Summary: identification and empirical design

- The S&P versus Moody’s comparison is central. Both sell ratings, but only S&P controls the S&P 500.
- Firm fixed effects compare the same firm over time.
- Quarter fixed effects absorb common market conditions.
- Published-rule controls absorb the criteria S&P says it uses.
- Vacancy and HighCAR tests show firms respond to index-entry incentives.
- The 2002 rule-change test provides an eligibility shock.
- Appendix robustness narrows alternative explanations: Fama-MacBeth, extended controls, FISC ratings, domicile definitions, and alternative timing all support the main result.

6 Tables and figures

6.1 Table 1 + Figure 1: The benchmark ecosystem becomes large

Read:

- Table 1 shows the growth of U.S. equity funds and index-tracking funds over 1980–2019.
- Panel A reports broad U.S. equity fund growth.
- Panel B focuses on S&P 500 funds: passive S&P 500 index funds grow from 1 fund in 1980 to 91 funds in 2019.
- Passive S&P 500 AUM grows from about \$0.1 billion in 1980 to about \$1.85 trillion in 2019.
- Other funds benchmarked to the S&P 500 also become large, reaching roughly \$1.83 trillion in AUM in 2019.
- Panel C provides a comparison with Russell 1000-related funds.
- Figure 1 presents the time series and makes the post-1990s acceleration visually clear.

Economic message: The financial value of index membership rises with the scale of assets tracking the index. This is the economic reason candidate firms may care about the committee’s decisions.

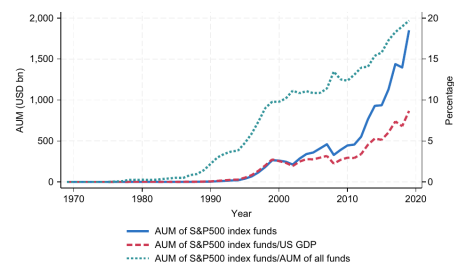
Table 1. Descriptive Statistics for the U.S. Equity Fund Market

Fund	1980	1990	2000	2010	2019
Panel A: All funds					
Total number of funds	237	727	2,425	2,543	2,608
(Total value in billions of USD)	(37.7)	(176.7)	(2,661.9)	(3,601.1)	(9,401.1)
Panel B: S&P 500					
Total number of S&P 500 index funds	1	13	97	82	91
(Total value in billions of USD)	(0.1)	(3.9)	(260.3)	(445.2)	(1,851.4)
Number of other funds benchmarking against S&P 500	96	210	714	590	481
(Total value in billions of USD)	(18.7)	(84.0)	(1,001.3)	(1,075.6)	(1,834.6)
Total number of funds benchmarking against S&P 500	97	223	811	672	572
(Total value in billions of USD)	(18.8)	(87.9)	(1,261.5)	(1,520.8)	(3,686.0)
Panel C: Russell 1000					
Total number of Russell 1000 index funds	0	0	3	10	40
(Total value in billions of USD)	(0.0)	(0.0)	(0.5)	(31.9)	(165.5)
Number of other funds benchmarking against Russell 1000	51	118	429	521	402
(Total value in billions of USD)	(10.7)	(51.7)	(737.9)	(608.6)	(1,175.9)
Total number of funds benchmarking against Russell 1000	51	118	432	531	442
(Total value in billions of USD)	(10.7)	(51.7)	(738.3)	(640.5)	(1,241.3)

Note: This table provides the descriptive statistics for the U.S. open-end equity funds and equity ETFs in various years from 1980 to 2019, based on data from Morningstar. The table shows the number of funds and assets under management (AUMs) at the end of a year. Panel B is for funds passively tracking S&P 500, other funds benchmarked to but not necessarily passively tracking the index, and the sum of the two. Panel C is similar for funds passively tracking or benchmarked to Russell 1000.

(a) Table 1: U.S. equity fund market

Figure 1. (Color online) Growth of S&P 500 Index Funds 1970–2019



Note: The graph shows a time series of the combined asset under management (AUM) of all funds tracking S&P 500 (including both mutual funds and ETFs), its share in percentage of U.S. GDP, and its share in the total AUMs of all funds.

(b) Figure 1: growth of S&P 500 index funds

6.2 Table 2 + Table 3: How much discretion is there?

Read:

- Table 2 reports 773 additions during 1980–2018.
- Only 80.47% of added firms satisfy all coded criteria.
- The share meeting all criteria varies by decade: 79.60% in the 1980s, 88.57% in the 1990s, 78.90% in the 2000s, and 75.00% in the 2010s.
- Criteria most often violated include financial viability and liquidity conditions.
- The existence of 6,320 eligible-but-not-added firms shows that rule satisfaction is not sufficient for entry.
- During 2015–2018 alone, 297 firms satisfy all conditions but are excluded.
- Table 3 shows that the main sample has 479,101 firm-quarter observations.
- The mean of *SP_Add* is 0.155, which reflects the rarity of entry.

- The mean of *Purchase_SP* is 0.088 and *Purchase_Any* is 0.103, so rating purchases are also uncommon but not negligible.

Economic message: The central institutional fact is that published criteria are necessary to understand S&P 500 additions but not sufficient to explain them. This creates the empirical space in which conflicts of interest can operate.

Table 2. Rules vs. Discretion in S&P 500 Additions

	1980s	1990s	2000s	2010s	1980–2018	2015–2018
Panel A: Number of additions						
	201	175	237	160	773	92
Panel B: Percentage of stocks meeting a given criterion						
Meet all criteria	79.60	88.57	78.90	75.00	80.47	78.26
MinCap $\geq$ S&P 500 threshold	100.00	100.00	98.31	97.30	98.97	97.83
Turnover ≥ 1	100.00	100.00	99.58	99.38	99.74	98.91
Monthly volume $\geq$ 250,000 shares	82.09	97.71	97.89	99.38	94.05	100.00
Earnings_Last1Q > 0	98.51	94.86	91.56	93.12	94.44	90.22
Earnings_Last4Q > 0	97.01	90.86	90.72	95.00	93.27	93.48
IWF $\geq$ required threshold	100.00	100.00	99.58	98.75	99.61	97.83
Domicile_OK	100.00	100.00	93.25	83.12	94.44	91.30
Panel C: Number of (unique) firms that satisfied all addition criteria but not added to the index						
	1,780	4,230	3,552	438	6,320	297

Notes. This table reports the number of stocks added to S&P 500 (panel A), the fraction of additions that meet a given criterion (panel B), and the number of firms that satisfy all criteria but not added to the index (panel C), respectively. Note that the domicile requirement starts from Q3 2002, and the IWF requirement starts from 2004. IWF refers to the Investable Weight Factor which is calculated as the ratio of available float shares over total shares outstanding. For periods prior to the relevant implementation date, we assume the criteria are satisfied.

(a) Table 2: rules versus discretion

Table 3. Sample Summary Statistics

Variables	Number of observations	Mean	Standard deviation	Median
<i>SP_Add</i>	479,101	0.155	3.935	0.000
<i>Purchase_SP</i>	479,101	0.088	0.283	0.000
<i>Purchase_Any</i>	479,101	0.103	0.304	0.000
<i>Domicile_OK</i>	479,101	0.940	0.238	1.000
<i>Log(MinCap)</i>	479,101	5.265	1.732	5.235
<i>MinCap $\geq$ S&P 500 threshold</i>	479,101	0.711	0.453	1.000
<i>Turnover</i>	479,101	1.420	1.637	0.835
<i>Turnover ≥ 1</i>	479,101	0.938	0.242	1.000
<i>Log(average monthly volume)</i>	479,101	13.678	2.410	13.819
<i>Monthly volume $\geq$ 250,000 shares</i>	479,101	0.612	0.487	1.000
<i>Earnings_Last1Q</i>	479,101	6.904	23.915	1.526
<i>Earnings_Last1Q > 0</i>	479,101	0.738	0.440	1.000
<i>Earnings_Last4Q</i>	479,101	25.257	84.641	5.814
<i>Earnings_Last4Q > 0</i>	479,101	0.732	0.443	1.000
<i>IWF $\geq$ required threshold</i>	479,101	0.987	0.112	1.000
<i>Difference in sectoral representation</i>	479,101	-0.003	0.014	-0.001
<i>S&P500/600</i>	479,101	0.187	0.390	0.000
<i>Acquirer of S&P 500 firm</i>	479,101	0.000	0.011	0.000

Notes. This table provides summary statistics for the variables used in an analysis of the relationship between rating purchases and S&P 500 addition probability. The sample period is from 1980 to 2018. The sample contains newly added S&P 500 firms and firms that are not added but meet the sample construction criteria.

(b) Table 3: summary statistics

6.3 Table 4 + Figure 2: Rating purchases and addition probability

Read:

- Table 4 directly tests whether new rating purchases predict addition to the S&P 500.
- Column (1), a logit specification, estimates a positive S&P purchase coefficient of 0.377.
- In OLS columns with controls and fixed effects, *Purchase_SP* is around 0.200–0.225 percentage points and statistically significant.
- The preferred OLS coefficient of 0.214 is large relative to the 0.155 percentage-point unconditional addition probability.
- *Purchase_Any* is not the main driver; it is small or insignificant in most columns.
- Column (5) shows the effect is largest among the most plausible candidates: *Purchase_SP* $\times$ *SizeRank*[1, 300] is 1.700, [301, 500] is 0.813, and [501, 700] is 0.212.
- Column (6) shows *Purchase_Both* and *Purchase_SPOnly* are positive, while *Purchase_MoodyOnly* is near zero.
- Column (7) shows *Purchase_Moody* is negative and insignificant, whereas *Purchase_Any* is positive when specified differently.
- Figure 2 shows that *Purchase_SP* receives high random-forest importance relative to many variables in the public methodology.

Economic message: The table says that S&P-specific fee-generating interactions predict index addition in a way that ordinary rating demand does not. The size-rank pattern is especially important because the effect is strongest where index entry is realistic.

6.4 Table 5 + Table 6: Do firms strategically buy more S&P ratings?

Read:

- Table 5 asks whether firms buy more ratings when an index vacancy is created by a merger between existing S&P 500 firms.

Table 4. Do Rating Purchases Enhance the Probability of Being Added to S&P 500?

Variables	SP_Add = 100, Otherwise = 0					
	Logit	OLS				
	(1)	(2)	(3)	(4)	(5)	(6)
Purchase_Any	-0.012 (0.214)	-0.044 (0.074)	-0.033 (0.074)	-0.009 (0.073)	0.018 (0.072)	0.180*** (0.063)
Purchase_SP	0.377* (0.218)	0.214** (0.083)	0.225*** (0.083)	0.200*** (0.083)	-0.021 (0.070)	
Purchase_Moody						-0.037 (0.078)
Purchase_Both					0.199*** (0.061)	
Purchase_SP Only					0.184*** (0.063)	
Purchase_Moody Only					-0.008 (0.073)	
Purchase_SP $\times$ Size rank[1,300]					1.700*** (0.621)	
Purchase_SP $\times$ Size rank[301,500]					0.813** (0.320)	
Purchase_SP $\times$ Size rank[501,700]					0.212* (0.108)	
Purchase_SP $\times$ Size rank[701,1000]					0.029 (0.038)	
Controls	✓	✓	✓	✓	✓	✓
Industry FE						
Firm FE						
Quarter FE						
Observation	479,101	479,101	479,101	479,101	479,101	479,101
R ²	0.31	0.01	0.01	0.06	0.07	0.06

Notes. This table investigates how firms' purchase of new ratings affects their probability of being added to the S&P 500. The sample contains all firms that meet the minimum eligibility conditions, excluding those already in the index at the time of an addition decision. Column (1) is a logit regression, whereas columns (2)–(7) use linear models. The dependent variable *SP_Add* is a binary variable that equals 100 if a firm is added to S&P 500 in quarter t , and zero otherwise. The key independent variable, *Purchase_SP*, is a dummy variable that equals one if firm i has purchased new ratings from S&P in the current or any of the previous four quarters (i.e., over quarter $t-4$, t), and zero otherwise. *Purchase_Any* is a dummy variable that equals one if the firm has purchased any new ratings from either S&P or Moody's during the same period. In columns (6) and (7), we consider alternative model specifications. *Purchase_Both* is a dummy variable that equals one if firm i has purchased new ratings from both S&P and Moody's in quarter $t-4$, t , and zero otherwise. *Purchase_SP Only* (*Purchase_Moody's Only*) is a dummy variable that equals one if the firm has purchased any new ratings only from S&P (Moody's) during the same period. In column (7), *Purchase_Moody's* is a dummy variable that equals one if firm i has purchased new ratings from Moody's during the same period. The regression results showing the full set of controls are provided in Internet Appendix Table IA7. Industry, firm, and quarter fixed effects are included where indicated. Robust standard errors clustered at the firm and quarter levels are reported in parentheses. We include all the variables that reflect S&P 500 published methodology, including dummy variables of *MktCap* $\geq$ S&P 500 threshold, *Turnover* ≥ 1 , *Monthly volume* $\geq 250,000$ shares, *Earnings_Last1Q* > 0 , *Earnings_Last4Q* > 0 , and *IWF* $\geq$ required threshold as well as five interaction terms between *SP400/600* and *Turnover* ≥ 1 , *Monthly volume* $\geq 250,000$ shares, *Earnings_Last1Q* > 0 , *Earnings_Last4Q* > 0 , and *IWF* $\geq$ required threshold for the period after July 31, 2017. We also include the underlying continuous variables, including *Log(MktCap)*, *Turnover*, *Log(average monthly volume)*, and *Difference in sectoral representation*. In column (5), we also include the set of size rank dummies in the regression.

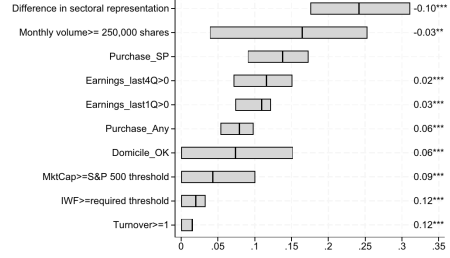
***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 4: rating purchases and addition probability

- In the full sample, $SPmerger \times SizeRank[1, 1000]$ is 0.012 for S&P and 0.005 for Moody's; the difference is 0.007.
- In the restricted sample, the analogous coefficients are 0.021 for S&P and 0.008 for Moody's; the difference is 0.013.
- When size buckets are separated, the largest differences appear among plausible entrants, especially ranks 301–500 and 501–700.
- Table 6 asks whether firms buy more S&P ratings when the payoff to index entry is high.
- For the full-sample top-1,000 group, the $SPmerger \times SizeRank[1, 1000] \times HighCAR$ coefficient is 0.016 for S&P and -0.003 for Moody's.
- In the restricted sample, the analogous coefficients are 0.022 for S&P and -0.011 for Moody's.
- The positive S&P and negative/near-zero Moody's pattern repeats across several plausible size groups.

Economic message: Tables 5 and 6 are mechanism tests. They show that firms do not merely buy S&P ratings and then happen to be added; they appear to increase S&P purchases when index opportunities and payoffs increase.

Figure 2. Random Forest and Relative Feature Importance



Notes. This figure plots the average and interquartile range for relative feature importance using random forest. All factors are dummy variables.

Figure 2: random forest feature importance

Table 5. Do Firms Buy More S&P Ratings When There Are Vacancies in the S&P 500?

Variables	Full sample						Restricted sample					
	SP	Moody's	Difference	SP	Moody's	Difference	SP	Moody's	Difference	SP	Moody's	Difference
	(1)	(2)	(3) = (1) - (2)	(4)	(5)	(6) = (4) - (5)	(7)	(8)	(9) = (7) - (8)	(10)	(11)	(12) = (10) - (11)
$SPmerger \times Size\ rank[11,000] (\beta_1)$	0.012*** (0.003)	0.005*** (0.002)	0.007**				0.021*** (0.004)	0.008** (0.004)	0.013**			
$SPmerger \times Size\ rank[1,300] (\beta_2)$				0.017* (0.009)	0.009 (0.006)	0.008				0.043** (0.017)	0.026** (0.011)	0.017
$SPmerger \times Size\ rank[301,500] (\beta_3)$				0.018*** (0.005)	0.003 (0.003)	0.014***				0.030*** (0.008)	0.008 (0.009)	0.023**
$SPmerger \times Size\ rank[501,700] (\beta_4)$				0.014*** (0.005)	0.007** (0.003)	0.007				0.028*** (0.007)	0.007* (0.004)	0.021***
$SPmerger \times Size\ rank[7,011,000] (\beta_5)$				0.008*** (0.003)	0.004* (0.002)	0.004				0.013*** (0.005)	0.007 (0.004)	0.007
Bond	0.643*** (0.013)	0.613*** (0.015)	0.030	0.643*** (0.013)	0.613*** (0.015)	0.030	0.585*** (0.018)	0.555*** (0.020)	0.030	0.584*** (0.018)	0.555*** (0.020)	0.030
Controls	✓			✓			✓			✓		
Firm-rating agency FE	✓			✓			✓			✓		
Quarter-rating agency FE	✓			✓			✓			✓		
Observation	1,063,672			1,063,672			372,296			372,296		
R ²	0.33			0.33			0.32			0.32		

Table 5: rating purchases around S&P 500 vacancies

Table 6. Do Firms Buy More Ratings When the Payoff for Entering the Index Is Higher?

Variables	Full sample						Restricted sample					
	SP	Moody's	Difference	SP	Moody's	Difference	SP	Moody's	Difference	SP	Moody's	Difference
	(1)	(2)	(3) = (1) - (2)	(4)	(5)	(6) = (4) - (5)	(7)	(8)	(9) = (7) - (8)	(10)	(11)	(12) = (10) - (11)
$SPmerger \times Size\ rank[11,000] \times HighCAR (\beta_1)$	0.016*** (0.005)	-0.003 (0.005)	0.019***				0.022*** (0.007)	-0.011 (0.009)	0.033***			
$SPmerger \times Size\ rank[11,000] (\gamma_1)$	0.007** (0.003)	0.006*** (0.002)	0.002				0.010* (0.006)	0.016** (0.008)	-0.006			
$SPmerger \times Size\ rank[1,300] \times HighCAR (\beta_2)$				0.036** (0.014)	-0.007 (0.016)	0.043**				0.000 (0.021)	-0.035 (0.023)	0.035
$SPmerger \times Size\ rank[1,300] (\gamma_2)$				0.008 (0.010)	0.010* (0.006)	-0.003				0.057** (0.022)	0.043** (0.019)	0.014
$SPmerger \times Size\ rank[301,500] \times HighCAR (\beta_3)$				0.028*** (0.009)	-0.006 (0.007)	0.034***				0.022* (0.013)	-0.034* (0.017)	0.056***
$SPmerger \times Size\ rank[301,500] (\gamma_3)$				0.011* (0.005)	0.005 (0.004)	0.006				0.022*** (0.008)	0.035** (0.016)	-0.013
$SPmerger \times Size\ rank[501,700] \times HighCAR (\beta_4)$				0.017* (0.009)	-0.005 (0.007)	0.022**				0.031** (0.013)	0.001 (0.009)	0.030**
$SPmerger \times Size\ rank[501,700] (\gamma_4)$				0.009* (0.005)	0.008** (0.003)	0.001				0.009 (0.009)	0.005 (0.005)	0.003
$SPmerger \times Size\ rank[7,011,000] \times HighCAR (\beta_5)$				0.009 (0.007)	0.000 (0.005)	0.009				0.020** (0.009)	-0.005 (0.009)	0.025**
$SPmerger \times Size\ rank[7,011,000] (\gamma_5)$				0.005 (0.003)	0.004 (0.002)	0.001				0.001 (0.007)	0.012 (0.008)	-0.011
Bond	0.643*** (0.013)	0.613*** (0.015)	0.030	0.643*** (0.013)	0.613*** (0.015)	0.030	0.584*** (0.018)	0.555*** (0.020)	0.030	0.584*** (0.018)	0.555*** (0.020)	0.029
Controls	✓			✓			✓			✓		
Firm-rating agency FE	✓			✓			✓			✓		
Quarter-rating agency FE	✓			✓			✓			✓		
Observation	1,063,672			1,063,672			372,296			372,296		
R ²	0.33			0.33			0.32			0.32		

Table 6: rating purchases when the inclusion payoff is higher

6.5 Table 7: 2002 foreign-firm rule change

Read:

- Table 7 uses the 2002Q3 rule change that made foreign firms ineligible for S&P 500 membership.
- The coefficient on $Post \times Foreign$ for S&P purchases is -0.007 and statistically significant.
- The same coefficient for Moody's purchases is 0.001 and insignificant.
- The S&P minus Moody's difference is -0.008 and statistically significant.

- In the size-rank decomposition, the strongest negative S&P effect appears for foreign firms ranked 301–500: -0.087.
- The table uses 68,522 observations and includes firm-rating-agency and quarter-rating-agency fixed effects.
- The paper interprets the baseline decline as a substantial reduction in foreign firms' S&P rating purchases after the index-membership channel disappears.

Economic message: This is the cleanest index-incentive shock. Foreign firms still may need credit ratings, but they lose the S&P 500 inclusion motive. The decline is S&P-specific, which is exactly what the governance hypothesis predicts.

Table 7. New Rating Purchases by Non-U.S. Firms after the 2002 Rule Change

	SP	Moody's	Difference	SP	Moody's	Difference
Variables	(1)	(2)	(3) = (1) – (2)	(4)	(5)	(6) = (4) – (5)
$Post \times Foreign (\beta_1)$	-0.007** (0.003)	0.001 (0.003)	-0.008***			
$Post \times Foreign \times Size\ rank[1,300] (\beta_2)$				0.005 (0.021)	0.073 (0.044)	-0.068***
$Post \times Foreign \times Size\ rank[301,500] (\beta_3)$				-0.087*** (0.022)	-0.016 (0.032)	-0.071***
$Post \times Foreign \times Size\ rank[501,700] (\beta_4)$				0.002 (0.051)	-0.012 (0.028)	0.014**
$Post \times Foreign \times Size\ rank[7,011,000] (\beta_5)$				-0.007 (0.023)	-0.015** (0.006)	0.009***
<i>Bond</i>	0.670*** (0.025)	0.609*** (0.023)	0.061*	0.670*** (0.025)	0.609*** (0.023)	0.061*
Controls	✓			✓		
Firm-rating agency FE	✓			✓		
Quarter-rating agency FE	✓			✓		
R^2	0.42			0.42		
Observation	68,522			68,522		

Notes. This table investigates how non-U.S. firms change their rating purchases, relative to U.S. firms, following the rule change in 2002Q3 on S&P 500 membership eligibility. We use a stacked framework in all regressions. The sample contains all firms listed on major U.S. stock exchanges that are not existing S&P 500 member firms during 2001Q3–2003Q3. *Post* is an indicator variable for 2002Q3 or later. *Foreign* is a dummy variable for non-U.S. firms listed on U.S. stock exchanges. Size rank dummies are defined the same as in Table 4. Other controls include *Log(MktCap)*, *M/B*, *Leverage*, *Profitability*, *o(ret)*, *Bond*, and *Missing controls*. Firm-rating agency and quarter-rating agency fixed effects are included. Robust standard errors clustered by firm-rating agency and quarter-rating agency are reported in parentheses. The regression results showing the full set of controls are provided in Internet Appendix Table IA10. In columns (3) and (6), we report the test statistics of the difference

Table 7: new rating purchases by non-U.S. firms after the 2002 rule change

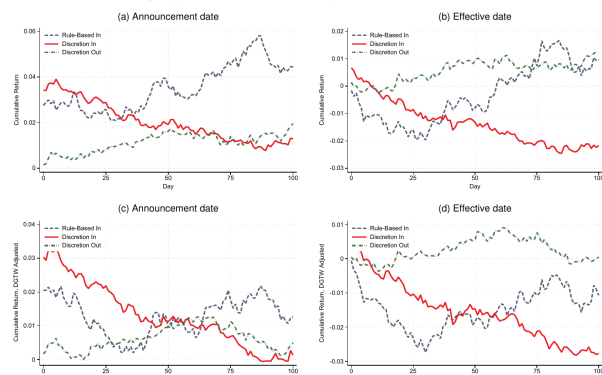
6.6 Figure 3 + Table 8: Are discretionary entrants better firms?

Read:

- Figure 3 compares stock performance of rule-based entrants, discretionary entrants, and discretionary outs.
- Discretionary entrants receive a positive initial announcement effect.
- Their performance fades after entry and is weaker over longer windows.
- Table 8 formalizes the comparison using announcement-date and effective-date windows.
- Discretionary entrants underperform rule-based entrants over longer post-inclusion windows.
- Discretionary entrants also underperform discretionary outs, which are eligible firms not selected despite looking stronger by the authors' criteria.
- The DGTW-adjusted returns tell a similar story to market-adjusted returns.

Economic message: If discretion were used to select firms with superior hidden quality, discretionary entrants should not systematically perform worse after inclusion. The performance evidence therefore undermines the benign-information explanation.

Figure 3. (Color online) Comparing Stock Performance: Discretionary Entrants and Other Stocks



Notes. This figure plots the average cumulative returns for three sets of stocks following their addition announcement dates and effective dates to S&P 500: rules-based entrants, discretionary entrants, and matched discretionary outs, respectively. (a) and (b) Market-adjusted returns.

(a) Figure 3: stock performance comparison

Table 8. Comparing Stock Price Performance: Discretionary Entrants vs. Other Stocks

	Discretionary entrants (1)	Rules-based entrants (2)	Discretionary outs (3)	Difference	
				(4) = (1) - (2)	(5) = (1) - (3)
Panel A: Announcement Date					
AD	0.034*** (0.001)	0.028*** (0.003)	0.002 (0.001)	0.007** (0.003)	0.033*** (0.002)
AD+20	0.031*** (0.004)	0.024** (0.010)	0.007* (0.004)	0.006 (0.011)	0.023*** (0.006)
AD+60	0.017** (0.007)	0.032** (0.016)	0.013 (0.008)	-0.016 (0.018)	0.004 (0.011)
AD+100	0.013 (0.009)	0.044** (0.020)	0.020* (0.010)	-0.032 (0.022)	-0.007 (0.014)
Panel B: Effective date					
ED	0.007*** (0.001)	-0.002 (0.002)	0.001 (0.001)	0.008*** (0.003)	0.005*** (0.002)
ED+20	-0.005 (0.004)	-0.015* (0.008)	0.003 (0.004)	0.010 (0.009)	-0.008 (0.006)
ED+60	-0.017** (0.007)	0.003 (0.016)	0.008 (0.008)	-0.020 (0.018)	-0.025** (0.011)
ED+100	-0.022** (0.009)	0.010 (0.020)	0.013 (0.010)	-0.031 (0.022)	-0.034** (0.014)
Panel C: Announcement date, DGTW					
AD	0.030*** (0.001)	0.021*** (0.002)	0.002** (0.001)	0.010*** (0.003)	0.029*** (0.002)
AD+20	0.022*** (0.004)	0.012 (0.009)	0.004 (0.003)	0.010 (0.010)	0.019*** (0.005)
AD+60	0.010 (0.006)	0.009 (0.014)	0.011** (0.005)	0.001 (0.015)	-0.001 (0.008)
AD+100	0.001 (0.008)	0.013 (0.017)	0.005 (0.006)	-0.011 (0.018)	-0.004 (0.010)
Panel D: Effective date, DGTW					
ED	0.006*** (0.001)	-0.001 (0.002)	0.000 (0.001)	0.007*** (0.002)	0.006*** (0.001)
ED+20	-0.006* (0.004)	-0.019*** (0.007)	0.001 (0.003)	0.013 (0.008)	-0.008* (0.005)
ED+60	-0.018*** (0.006)	-0.014 (0.014)	0.008 (0.005)	-0.004 (0.015)	-0.026*** (0.008)
ED+100	-0.028*** (0.008)	-0.011 (0.017)	0.000 (0.006)	-0.017 (0.019)	-0.028*** (0.010)

Notes. This table compares the relative stock price performance between discretionary entrants with both rules-based entrants and the matched discretionary outs. We report cumulative abnormal returns on the announcement date (AD) and effective date (ED), and over windows of 0, 20, 50, and 100 trading days from AD and ED, respectively. AD is the first trading day following the announcement, whereas ED is the first trading day after inclusion. Panels A and B are benchmarking to the market return, whereas panels C and D are adjusted returns using the DGTW return. The three groups of firms, that is, discretionary entrants, rules-based entrants, and discretionary outs, are defined dynamically following each addition event. To classify rules-based entrants and discretionary entrants, we use all eligible non-S&P 500 firms at the quarter of the addition announcement to select discretionary outs. *t*-Statistics on the differences between groups are reported in the last two columns. Standard errors are in the parentheses.

***, **, and * indicate statistically significant at the 1%, 5%, and 10% levels, respectively.

(b) Table 8: discretionary entrants versus other stocks

7 What the Internet Appendix adds

Appendix item	Role	Discussion
IA1	Leading S&P 500 funds	Names and sizes the benchmark ecosystem.
IA2	Market-cap thresholds	Clarifies rule coding over time.
IA3	S&P 500 vs Russell	Shows S&P 500 is less mechanical than Russell.
IA4–IA6	Rule-prediction tests	Membership is predictable; additions are much less so.
IA7, IA12–IA15, IA18, IA20, IA24	Addition-regression robustness	Confirms rating-purchase result across samples/specifications.
IA8–IA9	Vacancy and payoff robustness	Supports strategic purchase interpretation.
IA10	Foreign-firm rule change	Eligibility shock reduces purchases, not Moody's purchases.
IA16–IA17, Figures IA1–IA3	Performance and counterfactuals	Discretionary choices do not look value-enhancing.

8 Interpretation

8.1 What is the strongest evidence?

The strongest evidence is not any single table. It is the combination of five patterns:

1. published criteria leave meaningful discretion in addition decisions;
2. S&P rating purchases predict additions, while Moody’s purchases do not;
3. firms buy more S&P ratings when index vacancies appear;
4. the purchase response is stronger when index inclusion is more valuable;
5. firms that lose eligibility through the 2002 rule change reduce S&P purchases but not Moody’s purchases.

Together these patterns make a generic omitted-variable story less plausible than if the paper only had the baseline regression. The logic is especially strong because the paper uses Moody’s as a close comparison group. If rating purchases merely proxy for bond issuance, credit demand, firm maturity, or growth opportunities, then Moody’s purchases should behave similarly to S&P purchases. Instead, the evidence is concentrated in S&P purchases, which are tied to the same organization that controls the S&P 500.

This is also why the vacancy and rule-change tests matter. The baseline regression shows that S&P purchases predict additions. The vacancy test shows that firms buy more S&P ratings when the expected opportunity to enter the index rises. The high-CAR test shows that this response is stronger when the expected payoff to inclusion is larger. The 2002 foreign-firm rule change shows that when eligibility disappears, firms reduce S&P purchases. This sequence creates a revealed-preference interpretation: firms behave as if S&P ratings are part of the competition for S&P 500 membership.

8.2 What is the most important caveat?

The most important caveat is that the paper does not observe internal committee decisions, internal communications, fee negotiations, or explicit promises from S&P to firms. It therefore cannot prove that S&P explicitly rewards rating purchases. The paper shows that the patterns are consistent with conflict-of-interest concerns and hard to reconcile with a purely transparent rule-based system.

This caveat is similar to the broader conflict-of-interest literature in financial institutions. Many papers in that literature do not observe a direct “smoking gun” either; instead, they infer conflicts from systematic patterns in outcomes. For example, He, Qian, and Strahan (2012) show that large issuers receive more favorable ratings; Eling and Hau (2015) show that issuers giving more securitization business to rating agencies receive more favorable ratings; Baghai and Becker (2018) show that issuers buying nonrating services receive higher ratings; and Li, Lou, and Zhang (2024) show that commercial ties can bias ESG ratings. Li, Liu, and Wei apply a similar empirical logic, but to a different object: not the rating itself, but the index-membership decision.

A second caveat is that S&P may learn useful private information through the ratings process. In this benign interpretation, rating purchases help S&P identify firms that are better candidates for the index. The paper addresses this by comparing rules-based entrants, discretionary entrants, and discretionary outs. If the information channel improves index quality, discretionary entrants should outperform. Instead, discretionary entrants underperform rules-based entrants and matched discretionary outs after inclusion, which weakens the benign information interpretation.

8.3 Relation to index-addition literature

- The classic index-addition literature studies stock-price effects of being added to an index.
- This paper asks an earlier governance question: how are firms selected for addition in the first place?

- It also shows that not all additions are the same. Rules-based and discretionary additions have different post-inclusion performance.

The closest traditional literature studies whether index inclusion creates price pressure or changes stock behavior. Harris and Gurel (1986), Shleifer (1986), Beneish and Whaley (1996), Lynch and Mendenhall (1997), Wurgler and Zhuravskaya (2002), and Chen et al. (2004) document price effects around S&P 500 additions. Patel and Welch (2017) study longer-run returns and find partial reversals. These papers mainly treat index additions as events and ask what happens to prices after inclusion.

Li, Liu, and Wei move one step earlier. They ask whether the addition event itself is rule-based or discretionary. This distinction matters because index-addition studies often use additions as demand shocks. If addition decisions contain information or discretion, then the event is not a clean demand-only shock.

The paper also relates to the literature on index-induced comovement and style investing. Vijh (1994), Barberis and Shleifer (2003), Barberis, Shleifer, and Wurgler (2005), Greenwood (2007), and Chen et al. (2016) study how index membership or style classification can affect return comovement. In that literature, index inclusion changes how investors categorize the stock. Li, Liu, and Wei contribute by asking whether the categorization decision itself is potentially affected by commercial relationships.

The paper is also connected to the benchmarking literature. Kashyap et al. (2021) study the benchmark inclusion subsidy, and Kashyap et al. (2022) study whether there is too much benchmarking in asset management. These papers emphasize that benchmark membership has economic value. Li, Liu, and Wei build on that logic: if index membership is valuable, firms have incentives to compete for it through channels beyond ordinary firm fundamentals.

8.4 Relation to credit-rating literature

- Much credit-rating research studies rating inflation, issuer-pay incentives, and conflicts of interest in ratings themselves.
- This paper studies a different kind of conflict: a rating-agency business may interact with an index-provider business.
- The conflict is cross-product rather than within ratings alone.

The credit-rating literature usually studies whether rating agencies distort ratings because issuers are paying clients. He, Qian, and Strahan (2012) show that large issuers receive higher ratings. Eling and Hau (2015) show that securitization business can be associated with rating favors. Baghai and Becker (2018) show that issuers purchasing nonrating services receive higher ratings. Kedia, Rajgopal, and Zhou (2014) study changes in Moody's rating behavior after it became public. Li, Lou, and Zhang (2024) extend the conflict-of-interest logic to ESG ratings, showing that commercial ties may bias ESG-rating outcomes.

Li, Liu, and Wei's contribution is different. They do not primarily ask whether S&P gives inflated credit ratings to paying clients. Instead, they ask whether buying S&P ratings is associated with a separate product-market outcome: entry into an S&P-controlled index. The conflict is therefore organizational and cross-product:

$$\text{rating business relationship} \longrightarrow \text{index-membership decision.}$$

This is why Moody's is so important. Moody's is a close competitor in credit ratings but does not control the S&P 500. If the effect were only about credit-market activity, Moody's purchases should also predict additions. The fact that S&P purchases matter more supports the interpretation that the channel is linked to S&P's dual role.

8.5 Relation to conflict-of-interest literature in financial institutions

The paper also fits into the broader literature on conflicts of interest in financial institutions. Mehran and Stulz (2007) provide an early review of conflicts of interest in financial institutions. Ferreira et al. (2018) show that conflicts in bank-affiliated funds can affect performance. Fecht et al. (2018), Gil-Bazo et al. (2020),

and Haselmann et al. (2022) study how commercial ties and internal relationships can distort trading or investment behavior.

Li, Liu, and Wei extend this literature to index provision. In their setting, the possible conflict is not between a bank and its affiliated fund, or between a rating agency and a rating client, but between two business lines inside the same organization:

S&P as rating seller and S&P as index gatekeeper.

This is a new governance channel because the index provider controls a benchmark that reallocates capital across firms.

8.6 Relation to benchmark governance

- Benchmarks are often treated as neutral measurement devices.
- The paper shows that benchmark construction can have discretion and economic consequences.
- Transparency matters because researchers, funds, and regulators often rely on index membership as if it were objective.

This paper speaks directly to benchmark governance. Robertson (2023) argues that the S&P 500 should not be viewed as a purely passive index because more than 500 firms may satisfy eligibility conditions. Sharfman and Deluad (2022) argue for more disclosure by S&P and funds tracking the index, especially given discretionary decisions such as delayed Tesla inclusion and the exclusion of some share-class structures.

Li, Liu, and Wei go beyond these legal and institutional critiques by providing systematic empirical evidence. They do not simply count how many firms are large enough to qualify. Instead, they code S&P’s own published methodology, including financial viability, liquidity, public float, domicile, market capitalization, and sector representation. They also compare the S&P 500 with the Russell 1000. The appendix shows that Russell 1000 membership and additions are much more mechanically tied to size rankings, whereas S&P 500 additions are much less predictable from published rules. This comparison strengthens the benchmark-governance point: discretion is not inevitable for all indices; it is especially salient for the S&P 500.

8.7 Relation to post-addition performance literature

The paper also contributes to the literature on what happens after firms enter the index. Bennett et al. (2020) show that firms added to the S&P 500 underperform on average after addition, and they interpret this partly through a monitoring channel: index inclusion may reduce the share of active investors monitoring the firm. Sharfman and Deluad (2022) argue that discretionary index decisions can affect index performance.

Li, Liu, and Wei add a new distinction: discretionary entrants versus rules-based entrants. They find that discretionary entrants receive a short-run announcement effect, but then underperform rules-based entrants and discretionary outs after inclusion. This suggests that post-addition underperformance may not be a uniform feature of all index entrants. It may be concentrated among firms selected through discretion.

This distinction is important because it changes the interpretation of prior results. Instead of saying:

index addition $\Rightarrow$ post-addition underperformance,

the paper suggests:

discretionary addition $\Rightarrow$ post-addition underperformance.

8.8 Overall contribution relative to related literature

The paper combines three literatures that usually remain separate:

1. the index-addition literature, which studies price pressure, comovement, and index effects;

2. the credit-rating conflict literature, which studies issuer-pay incentives and commercial ties;
3. the benchmark-governance literature, which studies whether index construction is objective and transparent.

Its incremental contribution is to show that a major benchmark's membership decisions are not fully explained by the published methodology and that those discretionary decisions are systematically related to firms' purchases of S&P ratings. This creates a new form of potential conflict of interest: not rating inflation within the rating business, but a cross-business linkage between the rating business and the index business.

The central economic message is therefore:

valuable index membership + discretionary index selection
+ fee-based rating relationship = potential benchmark-governance problem.

9 Short summary

- The paper studies whether S&P 500 membership decisions are rule-based or discretionary.
- The S&P 500 matters because trillions of dollars track or benchmark to it.
- Firms benefit from inclusion through demand effects, visibility, prestige, and potentially lower cost of capital.
- S&P controls the index and also sells credit ratings, creating a potential conflict of interest.
- The authors code S&P's public methodology into observable rules.
- Published rules explain membership status better than they explain actual addition decisions.
- About 80.47% of additions satisfy all coded criteria, meaning roughly one in five additions involves at least one waiver or discretionary exception.
- Thousands of firms satisfy all criteria but are not added, showing that eligibility is not sufficient.
- Recent S&P rating purchases raise the probability of index addition in the baseline regression.
- Moody's purchases do not show the same relation.
- The S&P purchase effect is largest among firms with plausible size ranks for index entry.
- Random forests rank S&P purchases as important relative to several official methodology variables.
- Firms buy more S&P ratings when mergers between existing S&P 500 firms create vacancies.
- Firms buy more S&P ratings when the expected payoff from index inclusion is high.
- Foreign firms reduce S&P rating purchases after the 2002 rule change removes their eligibility, while Moody's purchases do not decline similarly.
- Discretionary entrants underperform rules-based entrants and matched discretionary outs after inclusion.
- The evidence does not prove direct intent or quid pro quo, but it raises serious governance questions.
- The broader message is that index construction can affect markets and should be treated as a governance issue, not merely a technical classification exercise.

10 Conclusion

10.1 Core conclusion

Li, Liu, and Wei document that S&P 500 membership decisions contain meaningful discretion and that this discretion is systematically related to firms' purchases of S&P credit ratings. The same pattern does not appear for Moody's ratings, even though Moody's is a close substitute in the rating market.

10.2 Economic intuition

The economic intuition is that S&P 500 membership is a scarce and valuable prize because index inclusion generates investor demand, stock-price gains, and potentially a lower cost of capital. Since S&P's published rules do not fully determine additions, firms may try to improve their chances through channels outside the public methodology. Because S&P also sells credit ratings, buying S&P ratings may create a commercial relationship, provide S&P with private information, or signal loyalty to the same organization that controls index inclusion. The fact that S&P rating purchases predict additions, Moody's purchases do not, firms buy more S&P ratings when index openings become available, and foreign firms reduce S&P purchases after losing eligibility all point to the same intuition: firms appear to treat S&P ratings as part of the competition for S&P 500 membership. The underperformance of discretionary entrants further suggests that discretion is not simply selecting better firms. The broader message is that benchmark construction can be economically consequential and vulnerable to conflicts of interest.

10.3 Incremental contribution

The paper's incremental contribution is to move beyond the traditional index-addition literature, which mainly studies price pressure, comovement, liquidity, and post-inclusion performance, and instead study the governance of the index-addition decision itself. Prior work recognizes that S&P 500 membership is not simply the largest 500 firms, but Li, Liu, and Wei measure discretion relative to S&P's own published methodology, including financial viability, liquidity, public float, domicile, and sector representation. They show that these rules explain membership much better than additions and that S&P 500 additions are far less mechanically predictable than Russell 1000 additions. The paper then introduces a new conflict-of-interest channel: firms' purchases of S&P ratings, but not Moody's ratings, are associated with higher index-addition probability. It further shows that firms buy more S&P ratings when index openings appear, when the payoff to inclusion is high, and that foreign firms reduce S&P purchases after losing eligibility in 2002. Finally, the paper reinterprets post-addition underperformance by showing that it is concentrated among discretionary entrants. Thus, the paper contributes to the credit-rating conflict literature, the index-addition literature, and the benchmark-governance literature by showing that index construction can be discretionary, commercially sensitive, and economically consequential.

10.4 Takeaway

Li, Liu, and Wei show that S&P 500 addition decisions involve economically meaningful discretion and that this discretion is systematically related to firms' purchases of S&P ratings. The Internet Appendix strengthens the result by documenting the robustness of the rating-purchase relation, contrasting S&P 500 discretion with Russell's more mechanical selection, and showing that discretionary entrants do not appear to outperform objective alternatives. The central takeaway is that benchmark construction is a governance problem, not only a measurement problem.

The paper's lasting contribution is to make index governance an empirical corporate-finance and market-design question. A benchmark can appear objective while still being shaped by opaque discretion. Because the S&P 500 is central to investment, research, and risk measurement, even small deviations from transparent rule-based selection can have large economic significance.